# Quarterly AI-human comparisons

**Fresh tests of the average quarterly gap, AI minus human** | 100 quarters, 1999Q1-2023Q4

| Variable | h | Mean gap | Adj. p | Median gap | Adj. p | SD gap | Adj. p |
|---|---|---|---|---|---|---|---|
| CPI | 0 | 0.090 | 1.000 | 0.047 | 1.000 | -0.719 | <0.001 |
| CPI | 1 | 0.177 | 1.000 | 0.164 | 1.000 | -0.584 | <0.001 |
| CPI | 4 | 0.062 | 1.000 | 0.074 | 1.000 | -0.437 | <0.001 |
| UNEMP | 0 | 0.026 | 1.000 | 0.036 | 1.000 | -0.118 | <0.001 |
| UNEMP | 1 | -0.030 | 1.000 | -0.010 | 1.000 | -0.190 | <0.001 |
| UNEMP | 4 | -0.101 | 1.000 | -0.082 | 1.000 | -0.311 | <0.001 |
| TBILL | 0 | -0.006 | 1.000 | 0.008 | 1.000 | -0.080 | <0.001 |
| TBILL | 1 | -0.048 | 1.000 | -0.031 | 1.000 | -0.150 | <0.001 |
| TBILL | 4 | -0.215 | 0.085 | -0.200 | 0.129 | -0.362 | <0.001 |
| RGDP | 0 | -113.011 | 0.070 | -136.938 | 0.008 | 59.644 | 0.030 |
| RGDP | 1 | -149.959 | 0.003 | -179.980 | <0.001 | 49.346 | 0.304 |
| RGDP | 4 | -246.126 | <0.001 | -289.686 | <0.001 | 40.839 | 1.000 |

Each gap averages the 100 differences between the AI and human quarterly mean, median, or sample SD. Adjusted p-values use a two-sided normal HAC test (Bartlett weights, 4 lags, finite-sample correction), with Holm adjustment across all 36 tests. These test average moment gaps, not equality of entire distributions. Gaps use each variable's native units: percentage points for CPI, UNEMP and TBILL; billions of chained dollars for RGDP.

## ARCHIVED ORIGINAL RESULTS: pages 2-4

The following tables reproduce all 483 saved moment differences and significance markers exactly as printed. **The historical input version is unresolved; these are not fresh validated tests.** The supplied original code pools rows across quarters. Its many-to-many availability join duplicates AI rows, altering their weights. In each of 1,000 simulations it independently assigns every pooled row an AI or human label with replacement; group sizes can change and quarter/person dependence is not preserved.

Archived stars: *** p <= 0.01; ** p <= 0.05; * p <= 0.10. Exact p-values are unavailable in the saved tables. No stars means the reported historical test was not significant at 10%. Printed negative zeros and rounded zeros with stars are preserved. Median, percentile, mean and SD differences use native variable units; skewness and excess-kurtosis differences are dimensionless. Source method: Simulating_SPF_processing/functions/tab_rand_test_moments.R.

**How to read the correlation plots and files**
Medians use the untrimmed source series (the official human SPF median); SDs are recomputed from individual responses with the sample denominator n-1. Zero SD quarters and duplicate AI response rows are retained. The solid line is OLS AI on human; the band is a pointwise 95% HAC(4) interval. The equality p-value tests intercept = 0 and slope = 1 jointly, using a chi-square reference. Correlations are descriptive, not equality tests. The CSVs include HAC(8) sensitivity, Holm adjustments, and first-difference correlations. Numerical p=0 in the correlation CSV denotes underflow; the finite log-p column retains its magnitude. RGDP uses changing survey-vintage base years; shared trends, vintage differences and influential observations affect interpretation. Current-data GDP slopes differ from the supplied example image. Source filenames and hashes are in the archival CSV.

# ARCHIVED ORIGINAL RESULTS

Current quarter (h = 0) | AI minus human pooled moment differences

| Variable | Median | P25 | P75 | Mean | SD | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|
| Nominal GDP | -355.21*** | -187.13*** | -632.35*** | -456.53*** | -410.48*** | -0.07*** | -0.10*** |
| GDP Price | -0.06** | -0.10*** | -0.19*** | -0.20*** | 0.25*** | 0.11*** | 0.70*** |
| Corporate Profits | -15.60*** | 159.80*** | -26.04*** | 15.81*** | -97.49*** | -0.09*** | -0.04*** |
| Unemployment Rate | 0.02** | 0.00 | 0.24*** | 0.03*** | -0.04*** | -0.47*** | -3.91*** |
| Non-Farm Payroll | -273.08*** | -249.45*** | -1626.00*** | -852.83*** | -706.25*** | 0.05*** | -0.07*** |
| Industrial Production | -1.58*** | -0.70*** | -1.30*** | -1.34*** | -0.47*** | 0.18*** | 0.58*** |
| Housing Starts | -0.03*** | 0.00 | 0.02*** | -0.00* | 0.02*** | 0.11*** | -0.06*** |
| Treasury Bill Rate (3M) | 0.13** | -0.04*** | -0.00*** | -0.01 | -0.01*** | -0.01 | 0.01 |
| AAA Corp Bond Yield | -0.17** | -0.05*** | -0.07** | -0.13*** | 0.02*** | 0.03*** | 0.08*** |
| BAA Corp Bond Yield | -0.02*** | -0.06*** | -0.03*** | -0.06*** | 0.01*** | -0.18*** | -0.16*** |
| Treasury Bond Rate (10Y) | 0.01*** | 0.04*** | 0.00 | 0.03*** | 0.03*** | 0.03*** | 0.01 |
| Real GDP | -125.70*** | -0.75 | -355.87*** | -350.34*** | -326.30*** | -0.13*** | -0.29*** |
| Real PCE | -49.48*** | 71.06 | -59.91*** | -120.64*** | -142.59*** | -0.11*** | -0.17*** |
| Real Non-Res Fixed Inv | -9.92*** | -6.75*** | 9.58 | -16.14*** | -34.06*** | -0.07*** | -0.23*** |
| Real Res Fixed Inv | -4.59*** | -0.70*** | -0.60*** | -0.75 | -4.44*** | 0.00 | 0.67*** |
| Real Federal C&GI | -5.66*** | -1.38*** | 0.67*** | 5.81*** | 20.45*** | 1.45*** | 10.40*** |
| Real State/Local C&GI | -9.14*** | -4.40*** | -7.30*** | -36.34*** | -16.99*** | -0.14*** | -0.25*** |
| Real Change in Private Inv | -0.90*** | -10.34*** | 7.80*** | -0.96*** | 14.62*** | 0.80*** | -10.83*** |
| Real Net Exports | 12.66*** | 5.00*** | 0.36*** | 5.28*** | 6.49*** | -0.17*** | 0.56*** |
| CPI Inflation Rate | 0.18*** | 0.30*** | 0.03* | 0.17*** | -0.03*** | 0.45*** | -0.63*** |
| Core CPI Inflation Rate | 0.06*** | 0.07*** | 0.02*** | 0.09*** | 0.08*** | -0.18*** | -1.08*** |
| PCE Inflation Rate | 0.10*** | 0.18*** | 0.45*** | 0.32*** | 0.08*** | 0.76*** | -2.02*** |
| Core PCE Inflation Rate | 0.00 | 0.05*** | -0.07*** | 0.08*** | 0.13*** | 0.28*** | 0.28*** |

*** p <= 0.01; ** p <= 0.05; * p <= 0.10. Exact p-values unavailable. Historical input version unresolved; copied original results, not newly validated tests.

# ARCHIVED ORIGINAL RESULTS

One quarter ahead (h = 1) | AI minus human pooled moment differences

| Variable | Median | P25 | P75 | Mean | SD | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|
| Nominal GDP | -474.22*** | -253.23*** | -722.03*** | -508.05*** | -429.26*** | -0.07*** | -0.08*** |
| GDP Price | -0.17*** | -0.23*** | -0.21*** | -0.27*** | 0.16*** | 0.11*** | 0.64*** |
| Corporate Profits | -23.85*** | 168.07*** | -33.85*** | 13.57*** | -96.72*** | -0.08*** | -0.05*** |
| Unemployment Rate | -0.04*** | 0.00 | 0.10 | -0.03*** | -0.09*** | -0.67*** | -6.10*** |
| Non-Farm Payroll | -387.50*** | -203.71*** | -1874.09*** | -928.51*** | -673.93*** | 0.10*** | 0.08*** |
| Industrial Production | -1.74*** | -0.47*** | -1.76*** | -1.65*** | -0.55*** | 0.18*** | 0.60*** |
| Housing Starts | -0.02*** | -0.04*** | 0.05*** | 0.01*** | 0.05*** | 0.16*** | -0.14*** |
| Treasury Bill Rate (3M) | -0.05** | -0.05*** | -0.09*** | -0.04*** | -0.01*** | 0.01 | 0.01 |
| AAA Corp Bond Yield | -0.20** | -0.09*** | -0.25*** | -0.17*** | 0.01*** | 0.05*** | 0.06*** |
| BAA Corp Bond Yield | -0.08*** | -0.10*** | -0.12*** | -0.12*** | -0.02*** | -0.20*** | -0.44*** |
| Treasury Bond Rate (10Y) | 0.00 | -0.10*** | 0.00 | -0.03*** | 0.02*** | 0.03*** | -0.05*** |
| Real GDP | -100.20*** | -18.18 | -406.65*** | -386.19*** | -337.21*** | -0.13*** | -0.27*** |
| Real PCE | -85.90*** | 30.89 | -135.33*** | -150.12*** | -153.34*** | -0.11*** | -0.15*** |
| Real Non-Res Fixed Inv | -26.33*** | -19.81*** | -8.22** | -25.60*** | -35.42*** | -0.06*** | -0.20*** |
| Real Res Fixed Inv | -4.22* | 1.43*** | 0.68*** | 0.79* | -3.99*** | 0.05* | 0.99*** |
| Real Federal C&GI | -9.40*** | -2.47*** | -2.33*** | 4.02*** | 21.55*** | 1.49*** | 10.83*** |
| Real State/Local C&GI | -8.02*** | -8.94*** | -8.90*** | -37.96*** | -18.41*** | -0.14*** | -0.26*** |
| Real Change in Private Inv | -3.13*** | -14.21*** | 6.00*** | -2.42*** | 14.15*** | 0.80*** | -15.27*** |
| Real Net Exports | 18.00*** | 20.97*** | 2.68*** | 13.06*** | 3.16*** | -0.19*** | 0.63*** |
| CPI Inflation Rate | 0.20*** | 0.20*** | 0.30*** | 0.24*** | 0.15*** | 0.40*** | -1.35*** |
| Core CPI Inflation Rate | 0.00 | 0.06*** | -0.08*** | 0.11*** | 0.12*** | 0.51*** | -1.67*** |
| PCE Inflation Rate | 0.10*** | 0.00 | 0.50*** | 0.25*** | 0.20*** | 0.89*** | -5.25*** |
| Core PCE Inflation Rate | -0.05*** | 0.00 | -0.10*** | 0.10*** | 0.14*** | 0.86*** | 0.21*** |

*** p <= 0.01; ** p <= 0.05; * p <= 0.10. Exact p-values unavailable. Historical input version unresolved; copied original results, not newly validated tests.

# ARCHIVED ORIGINAL RESULTS

Four quarters ahead (h = 4) | AI minus human pooled moment differences

| Variable | Median | P25 | P75 | Mean | SD | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|
| Nominal GDP | -633.59*** | -349.81*** | -857.04*** | -668.44*** | -441.11*** | -0.06*** | -0.07*** |
| GDP Price | -0.26*** | -0.37*** | -0.59*** | -0.47*** | -0.05** | 0.11*** | 0.46*** |
| Corporate Profits | 18.02*** | 105.75*** | -71.24*** | -1.25 | -92.81*** | -0.08*** | -0.10*** |
| Unemployment Rate | -0.10 | -0.12*** | 0.28*** | -0.09*** | -0.11*** | -0.35*** | -1.56*** |
| Non-Farm Payroll | -765.75*** | -508.25*** | -1850.70*** | -855.07*** | 159.53*** | 0.71*** | 3.98*** |
| Industrial Production | -2.36*** | -1.30*** | -3.40*** | -2.73*** | -1.10*** | 0.16*** | 0.56*** |
| Housing Starts | -0.04*** | -0.08*** | 0.14*** | 0.02*** | 0.11*** | 0.32*** | -0.30*** |
| Treasury Bill Rate (3M) | -0.35*** | -0.31*** | 0.15*** | -0.17*** | 0.11*** | 0.15*** | -0.03*** |
| AAA Corp Bond Yield | -0.24*** | -0.12*** | -0.38*** | -0.26*** | -0.02*** | 0.10*** | 0.05*** |
| BAA Corp Bond Yield | -0.15*** | -0.13*** | -0.29*** | -0.23*** | -0.06*** | -0.16*** | -0.49*** |
| Treasury Bond Rate (10Y) | -0.32*** | -0.20*** | -0.11*** | -0.17*** | 0.01*** | 0.08*** | -0.16*** |
| Real GDP | -215.25*** | -167.19*** | -515.68*** | -499.40*** | -340.24*** | -0.10*** | -0.25*** |
| Real PCE | -170.82*** | -72.42*** | -253.58*** | -246.25*** | -163.35*** | -0.08*** | -0.14*** |
| Real Non-Res Fixed Inv | -76.93*** | -22.89*** | -69.62*** | -55.31*** | -34.65*** | -0.01 | -0.12*** |
| Real Res Fixed Inv | -8.92*** | -1.80*** | 3.72*** | -1.66*** | 2.76*** | 0.27*** | 2.35*** |
| Real Federal C&GI | 4.00*** | -1.00*** | -13.58*** | 1.34 | 25.05*** | 1.61*** | 11.99*** |
| Real State/Local C&GI | -20.70*** | -13.24*** | -23.04*** | -44.90*** | -21.75*** | -0.12*** | -0.25*** |
| Real Change in Private Inv | -1.00*** | -15.40*** | 7.00*** | -2.96*** | 16.79*** | 1.36*** | -28.10*** |
| Real Net Exports | 31.87*** | 47.80*** | 18.00*** | 33.75*** | -6.46*** | -0.27*** | 0.84*** |
| CPI Inflation Rate | 0.00*** | 0.07*** | 0.00 | 0.09*** | -0.10*** | 0.44*** | -2.65*** |
| Core CPI Inflation Rate | -0.05*** | 0.05*** | -0.11*** | 0.07*** | 0.08*** | 1.11*** | -3.96*** |
| PCE Inflation Rate | 0.00 | 0.04*** | 0.20*** | 0.12*** | 0.03*** | 1.13*** | -4.78*** |
| Core PCE Inflation Rate | -0.02*** | -0.05*** | -0.10*** | 0.04*** | 0.10*** | 1.30*** | -2.00*** |

*** p <= 0.01; ** p <= 0.05; * p <= 0.10. Exact p-values unavailable. Historical input version unresolved; copied original results, not newly validated tests.